

Prior-Fitted Functional Flows: In-Context Generative Models for Pharmacokinetics

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Contact: ojedamarin@uni-potsdam.de Code, model, data: github.com/cesarali/pff

Problem & Core Idea

Sparse cohorts, continuous trajectories

A pharmacokinetic study is a set of sparse, noisy, and irregularly sampled concentration–time series from a small cohort, $\mathcal{S} = \{\mathcal{D}^1, \dots, \mathcal{D}^J\}$. For a new individual, we want a **distribution over plausible concentration functions**—not just one point estimate—conditioned on the cohort and, optionally, a few early observations.

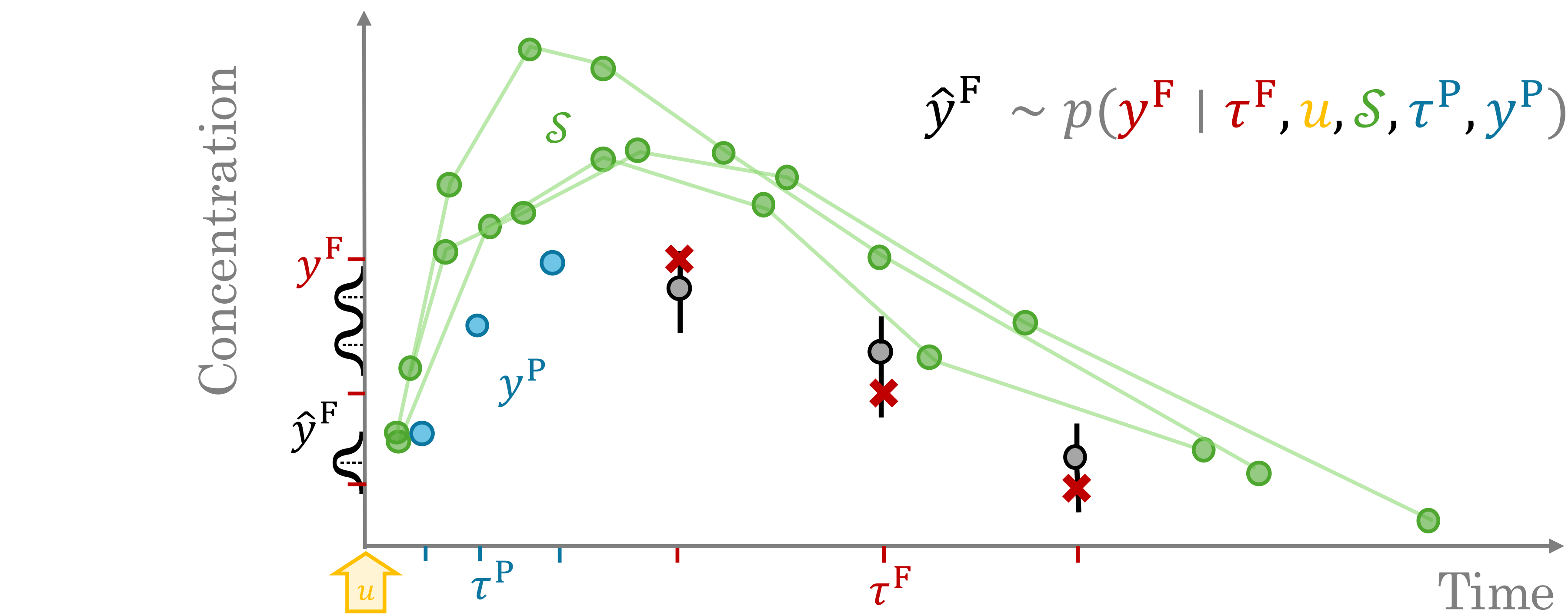


Figure: Study context $\mathcal{S}$ (green) and a patient's observed prefix (blue) define a distribution over future concentration values (gray/red).

Why prior fitting?

Classical nonlinear mixed-effects models require study-specific design, optimization, and diagnostics. High-capacity neural models are difficult to fit reliably to each small trial. **PFF** shifts that cost to a single pretraining stage:

- ▶ pretrain on diverse, mechanistically simulated PK studies;
- ▶ learn reusable inference directly in function space;
- ▶ deploy zero-shot to a new compound without fine-tuning.

Takeaway. One pretrained conditional flow supports both virtual-cohort generation and patient-level forecasting.

Physiologically Calibrated Prior

The simulator combines compartmental ODE systems with stochastic, time-varying parameters, diverse dosing routes and structures, and irregular sampling schedules. Its priors are calibrated against an open-access corpus of PK parameters mined from clinical bioequivalence studies.

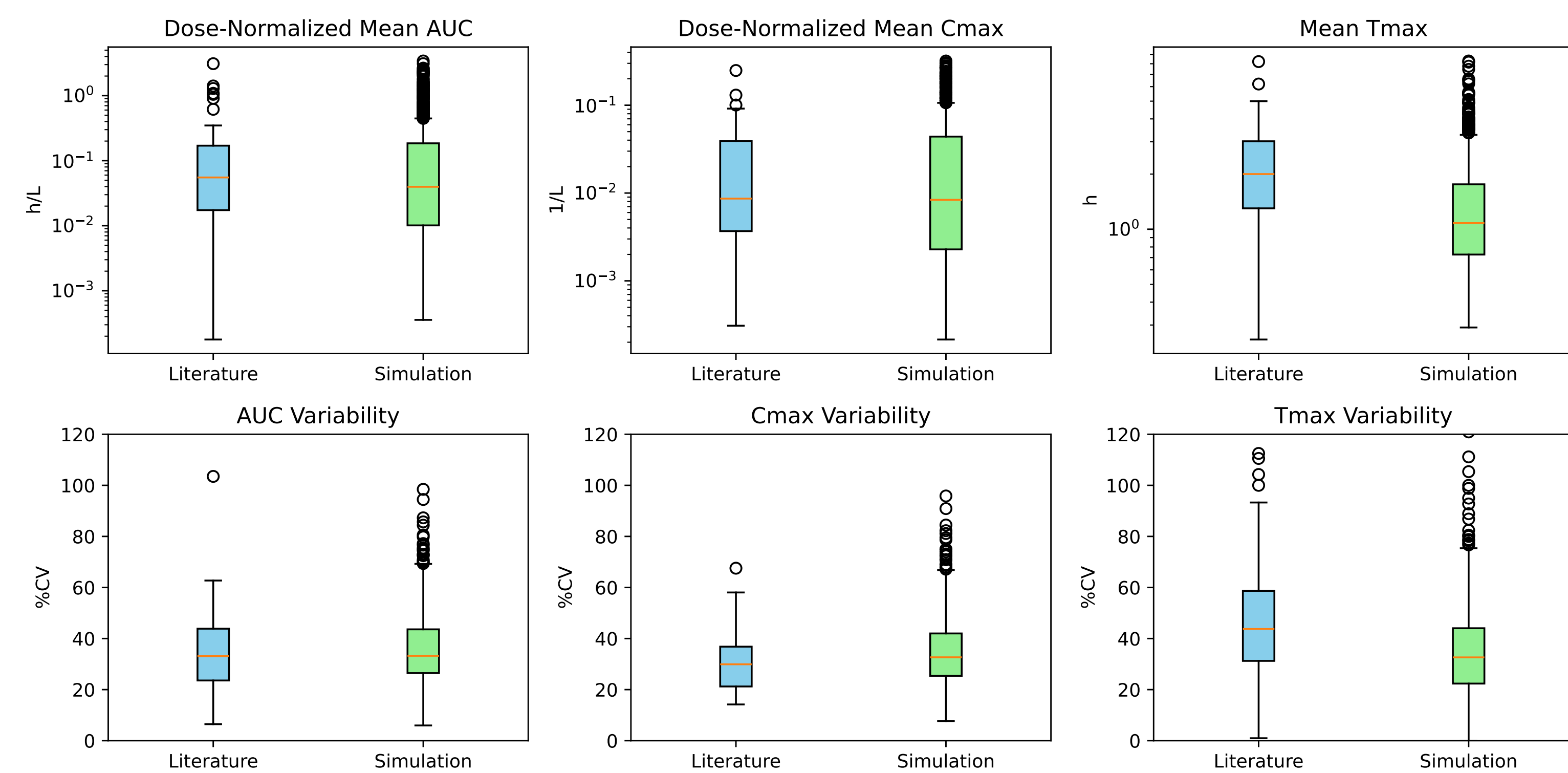


Figure: Literature (blue) and simulation (green) distributions for exposure, peak concentration, peak time, and their inter-individual variability.

Result. Literature calibration anchors the synthetic pretraining distribution to realistic scales and inter-individual variability.

Prior-Fitted Functional Flow

Conditional transport in function space

PFF learns a time-dependent neural-operator vector field that transports a tractable Gaussian-process (GP) reference measure into the conditional distribution of concentration functions induced by the PK simulator.

Population synthesis ($p = 0$)

Start from a GP prior over a full trajectory and transport it using study-level context $\mathcal{S}$.

The unified reference measure is

$$\eta_F(\cdot \mid o, p) = \begin{cases} \text{GP}_{\mathcal{T}_F}(\cdot \mid \mathbf{u}, \mathbf{y}_F), & p > 0, \\ \text{GP}_{\mathcal{T}_F}(\cdot), & p = 0. \end{cases}$$

For a sampled split, define $\mathbf{z}_0 = (\mathbf{y}_P, \mathbf{x}_F)$ and $\mathbf{z}_1 = (\mathbf{y}_F, \mathbf{y}_F)$. The Gaussian conditional path and masked flow-matching objective are

$$\mathbf{z}_t \sim \mathcal{N}(t\mathbf{z}_1 + (1-t)\mathbf{z}_0, \sigma_{\min}^2 I),$$

$$\mathcal{L}(\theta) = \mathbb{E} \left[\left\| \mathbf{M}_p \odot \mathbf{v}_t^\theta(\mathbf{z}_t, c) - (\mathbf{y}_F - \mathbf{x}_F) \right\|^2 \right].$$

The prefix-dependent mask $\mathbf{M}_p$ keeps the observed past fixed while the learned vector field transports uncertainty in the future.

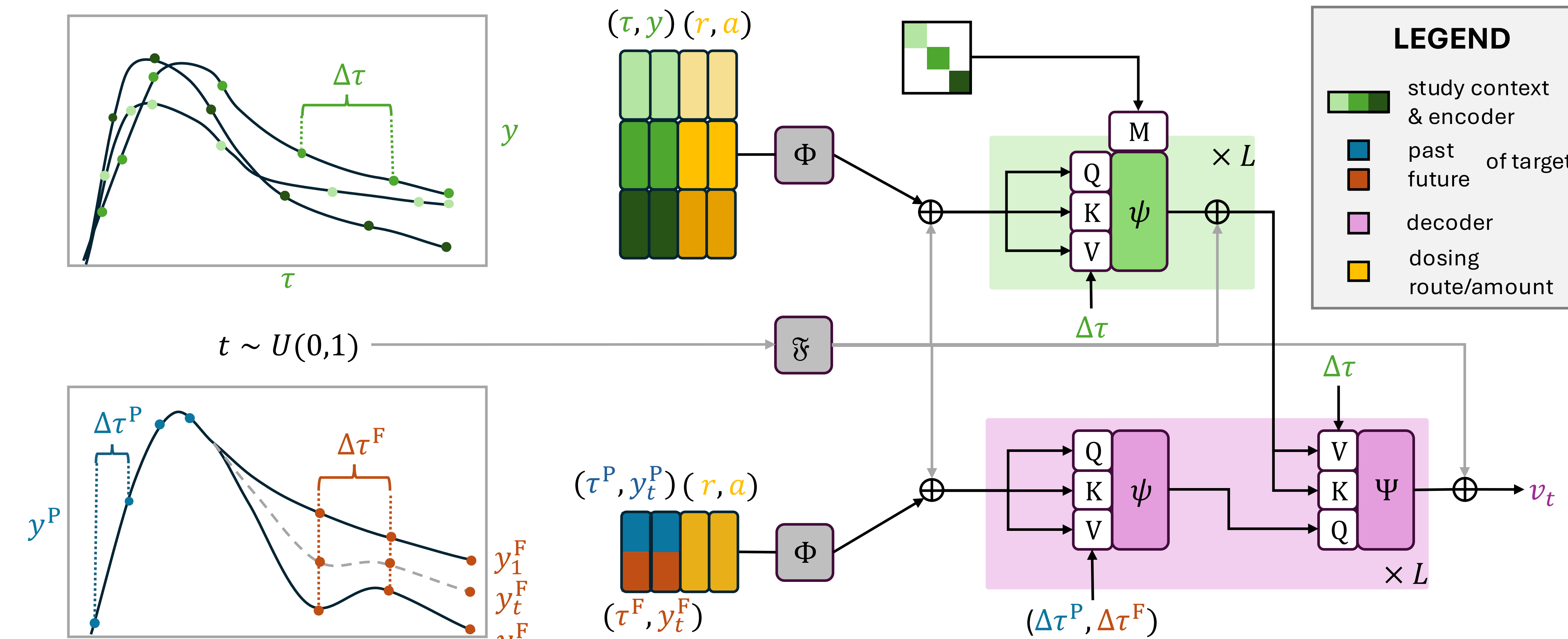


Figure: Operator encoder-decoder vector field. The context encoder summarizes individuals without mixing subjects; the target decoder combines the interpolated trajectory, dosing, flow time, and study context.

Continuum-attention neural operator

Observation grids differ across subjects and are not aligned. Encoder self-attention and decoder self-/cross-attention therefore use continuum-operator attention with trapezoidal quadrature corrections. This makes the learned vector field resolution- and grid-aware while retaining a transformer encoder-decoder structure.

Takeaway. GP posterior initialization enforces prefix consistency; triangular transport and continuum attention generate coherent futures on irregular time grids.

Zero-Shot Individual Forecasting

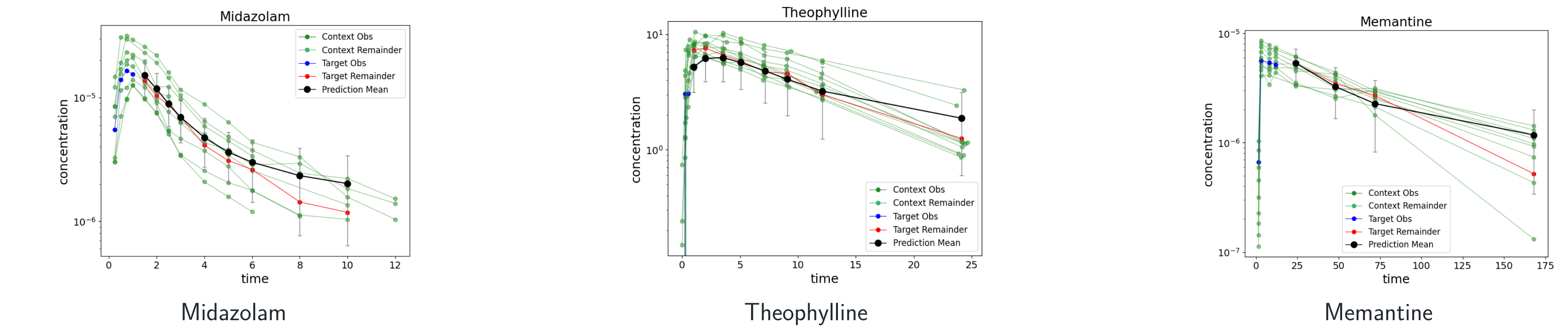


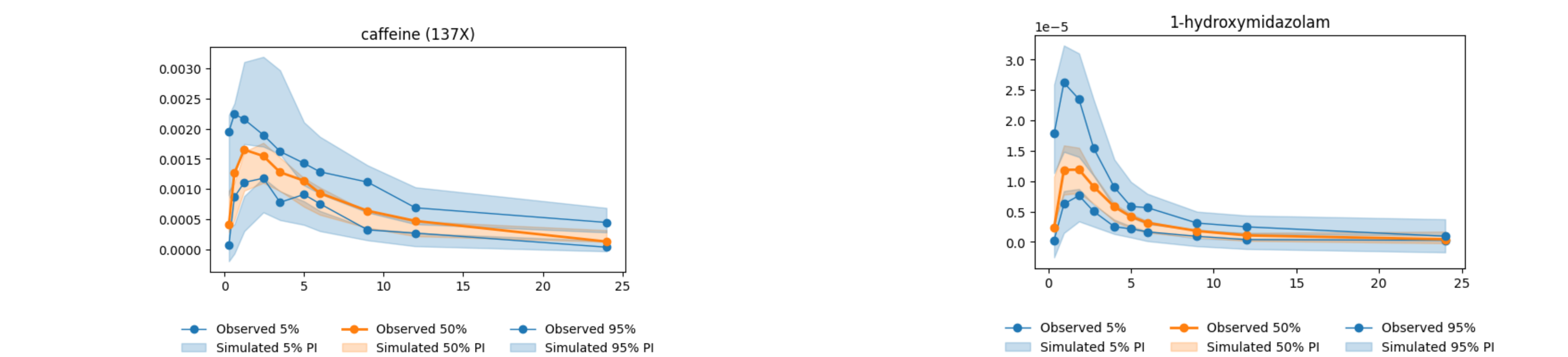
Figure: Conditioned on the cohort (green) and four early patient measurements (blue), PFF samples future trajectories (red); black denotes the predictive mean.

Selected Forecasting Accuracy (Log-RMSE)

Compound	NLME	Transf.	NODE	AICMET	PFF
caffeine	0.294	0.451	0.432	0.399	0.174
indometacin	0.582	0.429	0.350	0.536	0.245
omeprazole	1.300	0.842	1.003	0.936	0.593
theophylline	0.732	0.356	0.327	0.274	0.184
tolbutamide	0.723	0.678	0.578	0.517	0.421
1-OH-midazolam	–	0.567	0.737	0.491	0.422

Mean leave-one-individual-out log-RMSE. The table shows representative compounds from the paper on which PFF is best. Evaluation uses the first four observations as individual context.

Population Synthesis & Ablations



Caffeine

1-hydroxy-midazolam

Figure: Visual predictive checks: observed concentration summaries versus simulation-derived prediction intervals.

Synthetic evaluation	AUC, 0.5	MMD, 1.0
AICMET	0.54 ± 0.01	0.002 ± 0.001
PFF	0.52 ± 0.01	0.001 ± 0.001
PFF-standard attention	0.61 ± 0.03	0.004 ± 0.004
PFF-white noise	0.59 ± 0.05	0.004 ± 0.001

Generated trajectories are harder to distinguish from references than AICMET samples. Removing either the GP posterior or continuum attention degrades fidelity.

Sampling 100 virtual individuals takes about 3.5 s on one A100 GPU.

Result. PFF combines accurate forecasting with faithful zero-shot cohort synthesis; both the reference measure and operator architecture matter.